

Surface Energy and Microstructure in Coherent Phase Transitions

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Abstract

We study a variational problem involving a nonconvex function of ∇u , regularized by a higher order term. The motivation comes from the theory of martensitic phase transformation — specifically, a model for the fine scale structure of twinning near an austenite–twinned-martensite interface. It is widely believed that the fine scale structure can be understood variationally, through the minimization of elastic and surface energy. Our problem represents the essence of this minimization.

Similar variational problems have been considered by many authors in the materials science literature. They have always assumed, however, that the twinning should be essentially one-dimensional. This is in general false. Energy minimization can require a complex pattern of twin branching near the austenite interface. There are indications that the states of minimum energy may be asymptotically self-similar. ©1994 John Wiley & Sons., Inc.

1. Introduction

Solid-to-solid phase transitions often lead to fine-scale mixtures of distinct phases or phase variants with characteristic microgeometries. A case in point is martensitic transformation, which typically leads to twinning. There is a theory based on elastic energy minimization: see the work of Khachaturyan, Roitburd, and Shatalov, [15], [16], [17], [31], [32], for a geometrically linear treatment, and the work of Ball and James, [6], [7], for a geometrically nonlinear approach. The fine-scale structure arises because the elastic energy functional is not lower semicontinuous. Minimizing sequences tend to develop spatial oscillations, and these correspond physically to the observed fine-scale structure.

Elastic energy minimization successfully predicts many gross features of the microstructure, but not details such as its length scale. It is widely believed that such details may be captured by minimizing the sum of elastic and surface energy. Mathematically, surface energy can be represented as a singular perturbation involving strain gradients.

This paper analyzes a mathematical model for the fine-scale structure of an austenite–twinned-martensite interface in a crystal such as Indium-Thallium. The essence of the model is the minimization of elastic plus surface energy. A standard approach would be to use a “strain gradient theory” such as (1.1) below. In fact, it is more convenient to work with a related but more geometric model presented as (1.3) below. The model is derived in detail in [18]. The present paper is focused

primarily on the mathematical analysis of these variational problems. We refer to [18] for discussion of the physical implications of our results.

To get started, consider the problem of minimizing the strain-gradient type energy

$$(1.1) \quad E^\varepsilon(u) = \int_{R_L} u_x^2 + (u_y^2 - 1)^2 + \varepsilon^2 u_{yy}^2 dx dy$$

subject to

$$u|_{x=0} = 0,$$

where

$$R_L = (0, L) \times (0, 1).$$

The double-well potential $u_x^2 + (u_y^2 - 1)^2$ represents the elastic energy of the martensite: the preferred values $\nabla u = (0, \pm 1)$ are the stress-free states of two distinct variants of martensite. The “strain gradient” term $\varepsilon^2 u_{yy}^2$ represents surface energy. The boundary $x = 0$ corresponds to the austenite–twinned-martensite interface. We impose $u = 0$ there due to the constraint of elastic compatibility; see also (1.8) below.

Let us think for a moment about the case $\varepsilon = 0$, which corresponds to neglecting surface energy. It is easy to see that the minimum value of E^0 is zero, and that this minimum is not attained. A minimizing sequence can be constructed by taking u_h to be an oscillatory sawtooth function of y alone (with period h) in $(\delta, L) \times (0, 1)$, then using linear interpolation for $x \in (0, \delta)$ to satisfy the boundary condition.

As soon as $\varepsilon > 0$ the minimum of E^ε is attained. (The proof follows that of Theorem 2.1 below.) The higher order term $\varepsilon^2 u_{yy}^2$ limits the oscillations of u_y and thus establishes a length scale for the “microstructure.” Our goal is basically to understand the form of this microstructure, i.e., the character of the minimizer u_ε . We achieve this in an average sense rather than pointwise: our results are mainly concerned with scaling laws for the energy of u_ε .

It may seem curious that we have added a term in u_{yy} only, with no corresponding terms in u_{xx} or u_{xy} . The point is that the oscillations are primarily in u_y , and the rapid variation is mainly in the y direction, so a term in u_{yy} is sufficient to regularize the problem.

Prior to our work, it was widely believed that the minimizers of E^ε for small ε look more or less like the functions u_h described above (with the corners of the sawtooth smoothed out). Treatments based on this view will be found, for example, in [2], [6], [8], [13], [15], [16], [17], and [31]. A simple optimization in h and δ yields

$$\delta \sim \varepsilon^{1/2} L^{1/2}, \quad h \sim \varepsilon^{1/2} L^{1/2},$$

which leads to the scaling law $E^\varepsilon(u_h) \sim \varepsilon^{1/2} L^{1/2}$.

Our results indicate that this widely held belief is wrong. Indeed, it gives the *wrong scaling law* for the energy. The correct result is instead

$$\min_{u|_{x=0}=0} E^\varepsilon \sim C \varepsilon^{2/3} L^{1/3}.$$

To achieve this, one must take functions with relatively coarse oscillations for large values of x (where the boundary condition is hardly felt). The oscillations become finer as x decreases toward 0. We conjecture that for the optimal u , the refinement is asymptotically self-similar (as $\varepsilon \rightarrow 0$ and $x \rightarrow 0$ with $x/\varepsilon \rightarrow \infty$). Its length scale h should depend on the distance to the austenite x according to the law

$$(1.2) \quad h(x) \sim \varepsilon^{1/3} x^{2/3}.$$

See Theorems 2.6 and 3.5 below for precise statements of some partial results in this direction.

Actually, we shall not consider the functional E^ε directly. The reason is that a careful analysis of E^ε involves a lot of technical details — such as the smoothing of the sawtooth functions mentioned above — which are cumbersome and in our opinion not really essential. Instead, we shall work with a more “geometric” version of the problem. To formulate it, we observe that a minimizer u^ε of E^ε must have u_y^ε close to ± 1 except on a very small set. Moreover, following Modica (see [24]) and letting $H(t)$ be a primitive of $|t^2 - 1|$, we observe that

$$\begin{aligned} \int_{x=x_0}^2 (u_y^2 - 1)^2 + \varepsilon^2 u_{yy}^2 dy &\geq \int_{x=x_0}^2 2\varepsilon |u_y^2 - 1| |u_{yy}| dy \\ &= \int_{x=x_0}^2 2\varepsilon |H(u_y)_y| dy. \end{aligned}$$

The inequality becomes sharp if $\varepsilon u_{yy} = \pm(u_y^2 - 1)$, i.e., if in the layer where u_y changes from a value near -1 to a value near $+1$ we choose the right profile. Since u_y is essentially ± 1 , the variation of $H(u_y)$ is essentially $(H(1) - H(-1))/2$ times the variation of u_y . Absorbing constants into ε , we are thus led to consider the minimization of

$$(1.3) \quad I^\varepsilon(u) = \int_{R_L} u_x^2 + \varepsilon |u_{yy}| dx dy$$

subject to

$$|u_y| = 1 \text{ a.e.}, \quad u = 0 \text{ at } x = 0.$$

(The precise set of admissible functions is specified in Section 2.) This is basically a geometrical problem. The essential unknown is the set $\Omega^+ = \{(x, y) : u_y(x, y) = 1\}$. It determines u via integration in y (up to an arbitrary function of x). The surface energy is now $\varepsilon |u_{yy}|$: it counts the number of interfaces. We continue to view the term u_x^2 as representing the elastic energy of the martensite. Some people would call it a “non-local surface energy,” since it is determined in a non-local fashion by the phase geometry Ω^+ .

The functional I^ε has an interesting scaling property. If we set

$$u_\lambda(x, y) = \lambda u(\lambda^{-3/2} x, \lambda^{-1} y),$$

then u_λ is defined on $R_{L,\lambda} = (0, \lambda^{3/2}L) \times (0, \lambda)$, $|u_{\lambda y}| = 1$ a.e., and

$$(1.4) \quad I^\varepsilon(u_\lambda, R_{L,\lambda}) = \lambda^{3/2} I^\varepsilon(u, R_L).$$

By contrast E^ε has no such scaling invariance. It is partly for this reason that we prefer to deal with I^ε rather than with E^ε .

The passage from E^ε to I^ε may be viewed as removing a (mainly irrelevant) length scale of order ε from the problem (see [18] for further discussion). We shall not try to prove any rigorous link between E^ε and I^ε . Results of this kind for an analogous one-dimensional problem, however, will be found in [25] and [26].

The main result of this work is the following:

THEOREM 1.1. *The functional I^ε it attains its minimum, and there are constants $c, C > 0$ such that for sufficiently small ε*

$$(1.5) \quad c\varepsilon^{2/3}L^{1/3} \leq \min I^\varepsilon \leq C\varepsilon^{2/3}L^{1/3}.$$

Moreover, if u is a minimizer of I^ε then

$$(1.6) \quad c\varepsilon^{2/3}l^{1/3} \leq \int_{R_l} \varepsilon |u_{yy}| dx dy \leq C\varepsilon^{2/3}l^{1/3}$$

for every l , and a similar inequality holds for $\int_{R_l} u_x^2 dx dy$.

See Theorems 2.1 and 2.6 for more detail. The lower bound in (1.6) even holds in differentiated form, i.e., the integral over $x = l$ is bounded below by $c\varepsilon^{2/3}l^{-2/3}$. Note that this is a precise formulation of (1.2), since $\frac{1}{2} \int_0^1 |u_{yy}| dy$ is just the number of sign changes of u_y . In view of the scaling law (1.4) one might expect that for $Q_l = (0, l) \times (a, a + l^{2/3})$ one should have

$$(1.7) \quad I^\varepsilon(u^\varepsilon, Q_l) \leq C\varepsilon^{2/3}l.$$

So far we have been unable to prove or disprove this conjecture. We have, however, been able to prove an analogous but weaker result for minimizers of E^ε . It will be presented in [19].

Our results do not depend critically on the top and bottom boundary conditions. We could, for example, have used zero Dirichlet boundary conditions or periodic boundary conditions at $y = 0$ and $y = 1$.

We have said that the left-hand boundary condition $u = 0$ at $x = 0$ represents compatibility with the austenite. Actually, it corresponds to the extreme situation in which the austenite phase is completely rigid. More generally — and more realistically — one can consider the minimization of

$$(1.8) \quad \int_{R_L} (u_x^2 + \varepsilon |u_{yy}|) dx dy + \frac{\beta}{2\pi} \int_{-\infty}^0 \int_0^1 (u_x^2 + u_y^2) dx dy$$

subject to $|u_y| = 1$ a.e. in R_L . The parameter β represents an elastic modulus for the austenite. The minimization of the second integral amounts to solving $\Delta u = 0$ on

a half-strip; hence the second term can easily be expressed in terms of $u_0 = u|_{x=0}$ alone. Consider, for simplicity, the case of periodic boundary conditions in y ,

$$(1.9) \quad u(x, 1) = u(x, 0), \quad x \in (-\infty, L).$$

If we expand

$$u(0, y) = u_0(y) = \sum_{k=-\infty}^{\infty} \hat{u}_0 e^{2\pi i k y},$$

then the problem reduces after a brief calculation (see the Appendix) to minimizing

$$J^\varepsilon(u) = \int_{R_L} (u_x^2 + \varepsilon |u_{yy}|) dx dy + \beta \sum_{k=-\infty}^{\infty} |k| |\hat{u}_0(k)|^2$$

subject to $|u_y| = 1$. Note that the second term is proportional to the square of the $H^{1/2}/\mathbb{R}$ norm on $(0, 1)$. (See the Appendix for more information about $H^{1/2}$.) Many of the results stated above for the “hard” boundary condition $u_0 = 0$ can be extended to the case $\beta < \infty$; see Section 3 for detailed statements. We have not been able to obtain the lower bound in (1.6) for J^ε , but we can establish such a lower bound if the $H^{1/2}/\mathbb{R}$ norm of u_0 is replaced by a closely related quantity (see Theorem 3.5).

We digress slightly to make two observations. The first concerns the role of relaxation. The variational problem

$$\min_{|u_y|=1} \int_{R_L} u_x^2 dx dy$$

subject to given boundary conditions has, in general, no solution. The value of the minimum can be obtained, however, by solving the associated relaxed (convexified) problem

$$\min_{|u_y| \leq 1} \int_{R_L} u_x^2 dx dy,$$

with the same boundary conditions; see, e.g., [9] and [10]. Thus, to approximate the minimizers of I^ε one might reasonably start from a solution u_R of the relaxed problem, and “add suitable oscillations” (depending on u_R). We shall exploit this idea briefly in Lemma 2.5 below, as a method for constructing certain test functions. It remains unclear, however, how to connect the minimizers of I^ε with those of the relaxed problem in greater generality. (Some results in this direction will be presented in [19].)

Our second observation concerns the Modica-Mortola inequality mentioned above. Let $W : \mathbb{R}^n \rightarrow \mathbb{R}$ be a double-well potential (i.e., $W(a) = W(b) = 0$, $W > 0$ elsewhere) satisfying suitable growth and smoothness conditions. Then one can show (see [5], [12], [24], and [33]) that for minimizers v^ε of

$$(1.10) \quad \tilde{E}^\varepsilon(v) = \int_{\Omega} \frac{1}{\varepsilon} W(v) + \varepsilon |\nabla v|^2 dx$$

the inequality

$$(1.11) \quad \tilde{E}^\varepsilon(v^\varepsilon) \geq \int_{\Omega} 2W^{1/2}(v)|\nabla v| \, dx$$

is sharp as $\varepsilon \rightarrow 0$. The current work suggests that this is no longer true if v is itself required to be a gradient, $v = \nabla u$. Indeed taking $W(z) = z_1^2 + (z_2^2 - 1)^2$ and $\Omega = R_L$, our result on equipartition of energy suggests that

$$(1.12) \quad \int_{R_L} u_x^2 \, dxdy \approx \int_{R_L} (u_y^2 - 1)^2 + \varepsilon^2 u_{yy}^2 \, dxdy.$$

(See Lemma 2.4, and the beginning of its proof.) From the discussion above we expect

$$\int_{R_L} (u_y^2 - 1)^2 \, dxdy \approx \int_{R_L} \varepsilon^2 u_{yy}^2 \, dxdy \approx \int_{R_L} \varepsilon^2 |\nabla^2 u|^2 \, dxdy,$$

whence

$$\int_{R_L} W(\nabla u) \, dxdy \approx 3 \int_{R_L} \varepsilon^2 |\nabla^2 u|^2 \, dxdy.$$

This calculation suggests that

$$\begin{aligned} \int_{R_L} W(\nabla u) + \varepsilon^2 |\nabla^2 u|^2 \, dxdy &\approx \int_{R_L} \frac{2}{3} W(\nabla u) + 2\varepsilon^2 |\nabla^2 u|^2 \, dxdy \\ &\geq \int_{R_L} \frac{4}{\sqrt{3}} \varepsilon W^{1/2}(\nabla u) |\nabla^2 u| \, dxdy. \end{aligned}$$

Thus the elementary lower bound for E^ε analogous to (1.11) is not sharp in the limit as $\varepsilon \rightarrow 0$: it misses by a factor of $2/\sqrt{3}$. The hypothesis that we have approximate equality in (1.12) is slightly too optimistic. But the same argument applies (with $2/\sqrt{3}$ replaced by a different constant, again greater than 1) under the realistic hypothesis that the left hand side of (1.12) is bounded below by a constant times the right hand side.

Finally, let us mention some related work. Singularly perturbed problems of the type (1.10) have received a lot of attention recently; see, for example, [5], [12], [20], [24], [29], and [33]. Much less is known if v is itself a gradient, but see [3], [4], and [11] for some relevant results. The minimization of elastic plus surface energy has been widely studied in the material science literature, for example in [2], [8], [13], [15], [16], [31], and [32], but attention has mainly been focused on essentially one-dimensional constructions. The equations of viscoelastodynamics associated to the energy $\int u_x^2 + (u_y^2 - 1)^2$ have been studied numerically by Swart and Holmes in [34] and [35], using a finite difference scheme — which might possibly introduce a term of the type εu_{yy}^2 through “artificial viscosity”; their simulations show a few generations of twin branching, remarkably consistent with our self-similar construction. The problem discussed here is closely related to a ferromagnetic domain problem analyzed by Lifschitz in [21]. In that context Privorotskii found

a self-similar construction analogous to ours (see [30]), and Hubert obtained a scaling law analogous to (1.2), see [14].

2. The Model Energy with a Hard Boundary Condition

This section addresses the minimization of the model energy

$$I^\varepsilon(u) = \int_{R_L} u_x^2 + \varepsilon |u_{yy}| dx dy$$

subject to $|u_y| = 1$ a.e. and $u = 0$ at $x = 0$. The existence of a minimizer is not entirely trivial, because we have no terms in u_{xx} or u_{xy} ; it is proved using a suitable compactness argument (Theorem 2.1 and Lemma 2.2). To get a global upper bound, one must naturally find a suitable test function. This is done in Lemma 2.3, using a self-similar construction, leading to the conclusion that $\min I^\varepsilon \leq C\varepsilon^{2/3}L^{1/3}$. Deriving further properties of minimizers is naturally a more subtle matter: one must make some use of the first variation and/or introduce some local comparison functions. We use both techniques. The former leads to a crucial property we like to call “equipartition of energy”; it asserts that the relative contributions of the two terms u_x^2 and $\varepsilon |u_{yy}|$ are approximately equal, after integration in y only (see Lemma 2.4). We also use a local version of the self-similar construction (Lemma 2.5) to bound the energy in any rectangle $0 < x < l$, $0 < y < 1$. These are the two principal ingredients of our analysis; they lead to the *local* upper and lower bounds on the “energy” in any interval $0 < x < l$, which are our most important results.

The precise class of admissible functions for I^ε is

$$\mathcal{A}_0 = \{u \in H^1(R_L) : |u_y| = 1 \text{ a.e., } u_{yy} \text{ is a Radon measure on } R_L \text{ with finite mass, } u = 0 \text{ at } x = 0\}.$$

As usual, the spatial domain is $R_L = (0, L) \times (0, 1)$. We have taken “natural” boundary conditions at the top and bottom of R_L , but that is not essential. All the results in this section remain valid if one imposes periodicity, or even a Dirichlet condition $u = 0$ at $y = 0$ and $y = 1$.

Our first goal is existence:

THEOREM 2.1. *Assume that $\varepsilon > 0$. Then I^ε attains its minimum on $\mathcal{A}_0$.*

To prove Theorem 2.1 we require the following compactness result.

LEMMA 2.2. *Let $\Omega \subset \mathbb{R}^2$ be open and let $u^{(j)} : \Omega \rightarrow \mathbb{R}$ be a sequence such that $u^{(j)} \rightharpoonup u$ in $H^1(\Omega)$ and such that $u_{yy}^{(j)}$ lies in a compact subset of $H^{-1}(\Omega)$. Then*

$$u_y^{(j)} \rightarrow u_y \quad \text{in } L^2_{\text{loc}}(\Omega).$$

Proof: One proof makes use of compensated compactness methods (see [27] and [36]) applied to $v^{(j)} = (u_x^{(j)}, u_y^{(j)})$. Here we present instead a self-contained argument which was shown to us by L. Tartar. We may assume without loss of generality that $u = 0$. Let $\varphi \in C_0^\infty(\Omega)$. Then

$$-\int_{\Omega} \varphi^2 |u_y^{(j)}|^2 dx dy = \left\langle 2\varphi_y u_y^{(j)} + \varphi u_{yy}^{(j)}, \varphi u^{(j)} \right\rangle_{H^{-1}, H_0^1}.$$

Now (for a subsequence) $u_{yy}^{(j)} \rightarrow 0$ in H^{-1} by assumption and $u^{(j)} \rightarrow 0$ in L_{loc}^2 by the Sobolev embedding theorem. Hence $2\varphi_y u_y^{(j)} + \varphi u_{yy}^{(j)} \rightarrow 0$ in H^{-1} , $\varphi u^{(j)} \rightarrow 0$ in H_0^1 , and thus

$$(2.1) \quad \int_{\Omega} \varphi^2 |u_y^{(j)}|^2 dx dy \rightarrow 0.$$

The lemma follows.

Proof of Theorem 2.1: Fix $\varepsilon > 0$. Let $u^{(j)} \subset \mathcal{A}_0$ be a minimizing sequence for I^ε . Then $u^{(j)}$ is bounded in H^1 and $u_{yy}^{(j)}$ is bounded in $\mathcal{M}$ (the space of Radon measures). Passing to a subsequence one has

$$\begin{aligned} u^{(j)} &\rightharpoonup u && \text{in } H^1(R_L) \\ u_{yy}^{(j)} &\overset{*}{\rightharpoonup} u_{yy} && \text{in } \mathcal{M}. \end{aligned}$$

By standard lower semicontinuity results

$$I^\varepsilon(u) \leq \liminf_{j \rightarrow \infty} I^\varepsilon(u^{(j)}).$$

The only subtle point is to show that $u \in \mathcal{A}_0$. By assumption $u_y^{(j)}$ is bounded in L^∞ so that $u_{yy}^{(j)}$ is bounded in $\mathcal{M} \cap W^{-1, \infty}$. By Murat's lemma (see [28]) $u_{yy}^{(j)}$ lies in a compact set of H^{-1} . By the Lemma we have $u_y^{(j)} \rightarrow u_y$ in $L_{\text{loc}}^2(R_L)$. Passing to a further subsequence one sees that $|u_y| = 1$ a.e. We also have $u(0, y) = 0$ by standard trace theorems. Hence $u \in \mathcal{A}_0$ and the theorem is proved.

P. L. Lions has pointed out to us that pointwise convergence of $u_y^{(j)}$ can also be obtained by using the Aubin-Lions compactness lemma (see [22]), treating x as "time."

Our next goal is an upper bound for the minimum of I^ε . Here and throughout the rest of the paper, we always assume that ε is sufficiently small. More precisely (since ε has the dimensions of length), we assume that

$$(2.2) \quad \varepsilon^{1/3} L^{2/3} < c_0 < 1,$$

with the constant c_0 sufficiently small. This implies (as we shall see) that the number of "twins" — i.e., the number of sign changes of u_y — is large.

LEMMA 2.3. *There exists a constant C such that*

$$\min\{I^\varepsilon(u) : u \in \mathcal{A}_0\} \leq C\varepsilon^{2/3}L^{1/3}.$$

Proof: We subdivide the interval $0 < x < L$ geometrically as shown in Figure 1. The division points are $x_i = \theta^i L$, $i = 1, 2, 3, \dots$. We shall see later that θ should be chosen so that $\frac{1}{4} < \theta < \frac{1}{2}$.

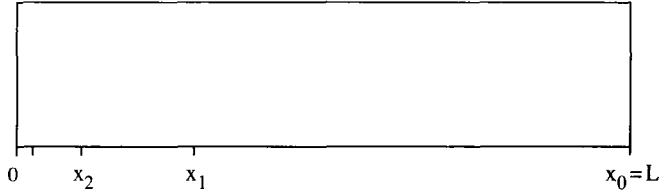


Figure 1. The subdivision points $x_i = \theta^i L$ in the proof of Lemma 2.3.

The main step will be to construct a function u on $[x_1, x_0] \times \mathbb{R}$ with the properties

$$(2.3) \quad |u_y| = 1 \quad \text{a.e.}$$

$$(2.4) \quad u(x, y+1) = u(x, y) \quad \text{for all } x, y$$

$$(2.5) \quad u(x_1, y) = \frac{1}{2}u(x_0, 2y) \quad \text{for all } y$$

and

$$(2.6) \quad \int_{x_1}^{x_0} \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx \leq C\varepsilon^{2/3}L^{1/3}.$$

Once this has been done, u can be extended to $(0, 1) \times \mathbb{R}$ by rescaling:

$$u(x, y) = 2^{-i}u(\theta^{-i}x, 2^i y) \quad \text{if } x \in [x_{i+1}, x_i].$$

It follows from (2.7) below that $u \in H^1(R_L)$. One easily verifies that $|u_y| = 1$ a.e., $u(0, y) = 0$, and

$$\begin{aligned} \int_{x_{i+1}}^{x_i} \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx &= \int_{x_1}^{x_0} \int_0^{2^i} [(2\theta)^{-2i}u_x^2 + 2^i \varepsilon |u_{yy}|] 2^{-i} \theta^i \, dy dx \\ &= \int_{x_1}^{x_0} \int_0^1 [(4\theta)^{-i}u_x^2 + (2\theta)^i \varepsilon |u_{yy}|] \, dy dx. \end{aligned}$$

If we choose θ to satisfy $\frac{1}{4} < \theta < \frac{1}{2}$, then it follows that

$$(2.7) \quad I^\varepsilon(u) = \sum_{i=0}^{\infty} \int_{x_{i+1}}^{x_i} \int_0^1 u_x^2 + \varepsilon |u_{yy}| dy dx \leq C \varepsilon^{2/3} L^{1/3}.$$

We note that when $\frac{1}{4\theta} = 2\theta$, i.e., when

$$\theta = 2^{-3/2},$$

the two terms $\int_0^1 u_x^2 dy$ and $\int_0^1 \varepsilon |u_{yy}| dy$ have the same scaling as $x \rightarrow 0$ (cf. Lemma 2.4 below).

It remains to construct the function $u : [x_1, x_0] \times \mathbb{R} \rightarrow \mathbb{R}$ satisfying (2.3) to (2.6). First, we define $v : [0, 1] \times [0, 1/2] \rightarrow \mathbb{R}$ by

$$v(x, y) = \begin{cases} y & \text{if } 0 \leq y \leq \frac{1}{8} + \frac{x}{8} \\ \frac{1}{4} + \frac{x}{4} - y & \text{if } \frac{1}{8} + \frac{x}{8} \leq y \leq \frac{3}{8} + \frac{x}{8} \\ -\frac{1}{2} + y & \text{if } \frac{3}{8} + \frac{x}{8} \leq y \leq \frac{1}{2} \end{cases}.$$

Then we extend v antiperiodically in y (see Figure 2).

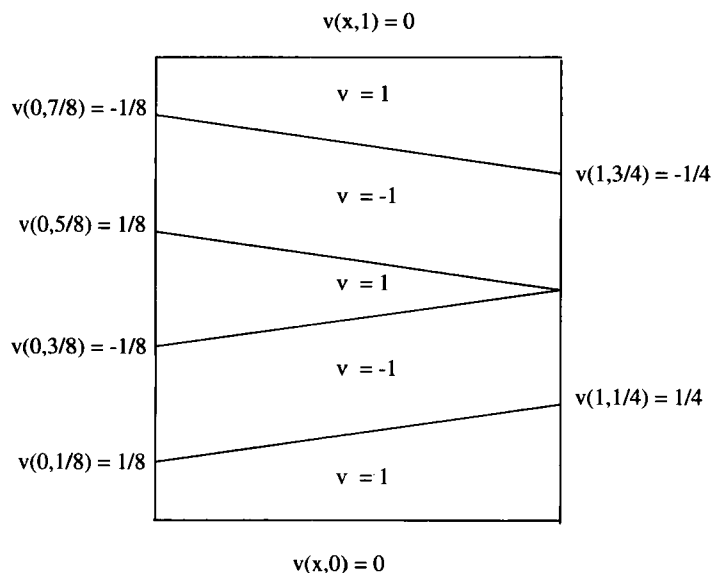
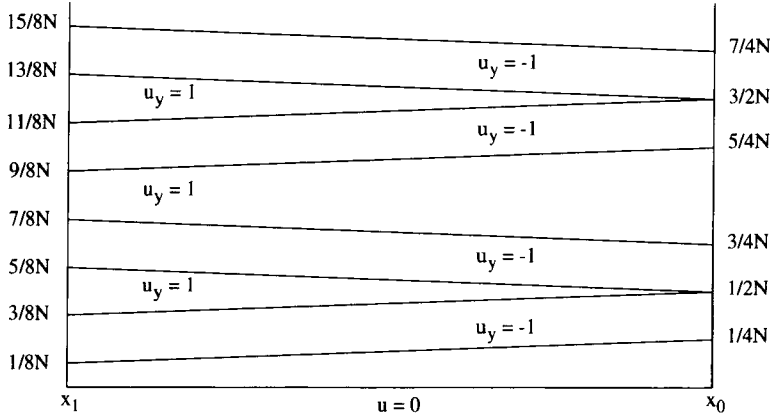


Figure 2. The function $v(x, y)$ in the proof of Lemma 2.3.

Figure 3. The function $u(x, y)$ in the proof of Lemma 2.3.

The resulting v satisfies

$$\begin{aligned}
 |v_y| &= 1 \quad \text{a.e.} \\
 v(x, y+1) &= v(x, y) \\
 v(0, y) &= \frac{1}{2}v(1, 2y) \\
 \int_0^1 \int_0^1 v_x^2 + \varepsilon |v_{yy}| \, dx dy &= \frac{1}{2} \left(\frac{1}{4} \right)^2 + 8\varepsilon.
 \end{aligned}$$

For $N \in \mathbb{N}$ let

$$u(x, y) = \frac{1}{N} v \left(\frac{x - x_1}{x_0 - x_1}, Ny \right)$$

(see Figure 3). Then clearly u satisfies (2.3) to (2.5). Moreover

$$\begin{aligned}
 \int_{x_1}^{x_0} \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx &= \frac{1}{32} \frac{1}{x_0 - x_1} \frac{1}{N^2} + (x_1 - x_0) 8\varepsilon N \\
 &= \frac{1}{32} \frac{1}{L(1 - \theta)} \frac{1}{N^2} + 8L(1 - \theta)N\varepsilon.
 \end{aligned}$$

If we choose $N \approx L^{-2/3} \varepsilon^{-1/3}$, then (2.6) follows. (Notice that N is large, on account of (2.2).)

We shall eventually show that the upper bound of Lemma 2.3 is within a constant of being sharp. The next result is a crucial step toward that goal. It shows that for a *minimizer* of I^ε , the two terms u_x^2 and $\varepsilon |u_{yy}|$ give nearly equal contributions to the energy.

LEMMA 2.4. (EQUIPARTITION OF ENERGY) *Let u be a minimizer of I^ε on $\mathcal{A}_0$. Then there exists a constant λ (depending on ε , L and u) such that*

$$(2.8) \quad \int_0^1 \varepsilon |u_{yy}|(x, y) dy - \int_0^1 u_x^2(x, y) dy = \lambda$$

for a.e. $x \in (0, L)$. The value of λ depends on u , and hence implicitly on ε and L , however it always satisfies

$$(2.9) \quad |\lambda| \leq C\varepsilon^{2/3}L^{-2/3}$$

with C independent of u , ε , and L . Furthermore, the trace $u^L(y) = u(L, y)$ satisfies $|u_y^L| = 1$ a.e. on $(0, 1)$, u_y^L is a Radon measure on $(0, 1)$ and

$$\int_0^1 \varepsilon |u_{yy}^L| dy = \lambda .$$

Finally, we have

$$\lim_{t \rightarrow 0} \frac{1}{t} \int_{L-t}^L \int_0^1 u_x^2 dy dx = 0 .$$

Proof: To explain why such results might be expected, we remark that similar assertions hold for minimizers of the “strain gradient” energy E^ε defined by (1.1). The proof is easy: the analogue of (2.8) is obtained by multiplying the Euler-Lagrange equation by u_x , then integrating in y . (This is basically the conservation law associated with the x -invariance of the integrand.) The assertions about u^L correspond to the natural boundary condition $u_x = 0$ at $x = L$.

We remark that the expression $\int_0^1 \varepsilon |u_{yy}|(x, y) dy$ in (2.8) is suggestive but slightly sloppy. We will show that the projection σ of $|u_{yy}|$ onto $(0, L)$ is absolutely continuous with respect to Lebesgue measure. The above integral then denotes the density of σ .

We now begin the proof in earnest. It exploits the fact that the integrand does not depend explicitly on x . For any $\varphi \in C_0^\infty(0, L)$, $(\text{Id} + t\varphi)$ is a diffeomorphism of $(0, L)$ onto itself for sufficiently small t . We let $\psi = (\text{Id} + t\varphi)^{-1}$ denote its inverse. Now for any minimizer u of I^ε , consider the variations defined by

$$u^t(x, y) = u(\psi^t(x), y) .$$

It is easy to verify that $|u_y^t| = 1$ a.e., $u^t|_{x=0} = 0$, and

$$(2.10) \quad \begin{aligned} \int_0^L \int_0^1 (u_x^t)^2(x, y) dy dx &= \int_0^L \int_0^1 u_x^2(\psi^t(x), y) [(\psi^t)']^2(x) dy dx \\ &= \int_0^L \int_0^1 u_x^2(x, y) dy \frac{1}{1 + t\varphi'(x)} dx . \end{aligned}$$

Formally one has similarly

$$\begin{aligned} \int_0^L \int_0^1 |u'_{yy}|(x, y) dy dx &= \int_0^L \int_0^1 |u_{yy}|(\psi^t(x), y) dy dx \\ &= \int_0^L \int_0^1 |u_{yy}(x, y)| dy (1 + t\varphi'(x)) dx, \end{aligned}$$

and one deduces (2.8) by (formally) differentiating with respect to t .

To make this argument rigorous takes a bit of work. Let $\Psi^t(x, y) = (\psi^t(x), y)$ and denote its inverse by $\hat{\Psi}^t$. Set $\mu = u_{yy}$ and define μ^t as the distributional derivative

$$\mu^t = u'_{yy} = (u_y \circ \Psi^t)_y.$$

As an initial step, we claim that

$$(2.11) \quad \langle \mu^t, \eta \rangle = \int_{R_L} (\eta \circ \hat{\Psi}^t) \det D\hat{\Psi}^t d\mu$$

for all $\eta \in C_0^1(R_L)$. Indeed,

$$\begin{aligned} \langle \mu^t, \eta \rangle &= - \int_{R_L} (u_y \circ \Psi^t) \eta_y dx dy \\ &= - \int_{R_L} u_y (\eta_y \circ \hat{\Psi}^t) \det D\hat{\Psi}^t dx dy. \end{aligned}$$

Now $\hat{\Psi}^t(x, y) = (x + t\varphi(x), y)$ and hence

$$(\eta_y \circ \hat{\Psi}^t) \det D\hat{\Psi}^t = [(\eta \circ \hat{\Psi}^t) \det D\hat{\Psi}^t]_y,$$

which implies (2.11).

It follows from (2.11) that μ^t is a Radon measure and hence that $u^t \in \mathcal{A}_0$. Splitting μ^t in its positive and negative parts we find that the total variation is given by

$$\begin{aligned} |\mu^t|(R_L) &= \int_{R_L} (\det D\hat{\Psi}^t) d|\mu| \\ (2.12) \quad &= \int_{R_L} (1 + t\varphi'(x)) d|\mu|. \end{aligned}$$

The projection σ of $|\mu|$ is defined by

$$\sigma(A) = |\mu|(A \times (0, 1)).$$

From (2.10) and (2.12) we see that $t \mapsto I^e(u^t)$ is differentiable at $t = 0$. Since u is minimizing we have

$$\begin{aligned} 0 &= \frac{d}{dt} \Big|_{t=0} I^e(u^t) \\ &= - \int_0^L \left(\int_0^1 u_x^2 dy \right) \varphi'(x) dx + \int_0^L \varphi'(x) d\sigma \end{aligned}$$

for all $\varphi \in C_0^\infty(0, L)$. It follows that σ is absolutely continuous with respect to Lebesgue measure, and writing $\sigma = \sigma(x) dx$ one has

$$\sigma(x) - \int_0^1 u_x^2(x, y) dy = \text{const} ,$$

which proves (2.8). To obtain (2.9) we simply integrate from 0 to L , then apply Lemma 2.3.

To establish the remaining parts of the lemma, we must consider variations that change the values at $x = L$. (The results at $x = L$ are presented for the sake of completeness only; they are not actually used in the sequel. The impatient reader is therefore advised to skip the rest of the proof.) Let

$$u'(x, y) = u((1-t)x, y) .$$

Since u is only defined for $x \leq L$ we assume for the moment that $t \geq 0$. Returning to our usual notation for the total variation of u_{yy} and arguing as above we find that

$$\begin{aligned} 0 &\leq I^\varepsilon(u') - I^\varepsilon(u) \\ &= \int_0^{(1-t)L} \int_0^1 \left[u_x^2(1-t) + \varepsilon |u_{yy}| \frac{1}{(1-t)} \right] dy dx \\ &\quad - \int_0^{(1-t)L} \int_0^1 u_x^2 + \varepsilon |u_{yy}| dy dx - \int_{(1-t)L}^L \int_0^1 u_x^2 + \varepsilon |u_{yy}| dy dx . \end{aligned}$$

We now substitute $\tau = tL$, divide by τ , and take the limit $\tau \rightarrow 0$. Making use of (2.8), we deduce that

$$(2.13) \quad \limsup_{\tau \searrow 0} \frac{1}{\tau} \int_{L-\tau}^L u_x^2 + \varepsilon |u_{yy}| dy dx \leq \lambda .$$

We shall use (2.13) to show that the trace u^L satisfies $|u_y^L| = 1$ a.e. and

$$\varepsilon \int_0^1 |u_{yy}^L| \leq \lambda .$$

To this end, let $t_n \rightarrow 0$ and define the rescaled functions

$$v^{(n)}(x, y) = u(L - t_n + xt_n, y) .$$

Note that $|v_y^{(n)}| = 1$ a.e. By (2.13) we have

$$(2.14) \quad \limsup_{n \rightarrow \infty} \int_{(0,1)^2} \varepsilon \left| v_{yy}^{(n)} \right| + t_n^{-2} \left(v_x^{(n)} \right)^2 dx dy \leq \lambda .$$

Here we have used the fact that the projection σ of $|u_{yy}|$ is of the form $\sigma(x)dx$ so that the change of variables formula applies. Notice that $v^{(n)}(1, y) = u(L, y) = u^L(y)$. By standard trace arguments

$$\int_{(0,1)^2} |v^{(n)} - u^L|^2 dx dy = \frac{1}{t_n} \int_0^1 \int_{L-t_n}^L |u - u^L|^2 dx dy \rightarrow 0$$

as $n \rightarrow \infty$. Hence

$$\begin{aligned} v^{(n)} &\rightharpoonup u^L \quad \text{in } H^1((0, 1)^2) \\ v_{yy}^{(n)} &\overset{*}{\rightharpoonup} u_{yy}^L \quad \text{in } \mathcal{M}((0, 1)^2). \end{aligned}$$

At first we only know this for a subsequence, but since the limit is unique the whole sequence must converge. Using Lemma 2.2 and Murat's Lemma as in the proof of Theorem 2.1 we deduce that $|u_y^L| = 1$ a.e. Furthermore, by the lower semicontinuity of the norm and (2.14), we have

$$\begin{aligned} \int_0^1 \varepsilon |u_{yy}^L| dy &= \int_{(0,1)^2} \varepsilon |u_{yy}^L| dy dx \\ (2.15) \quad &\leq \lambda - \liminf_{n \rightarrow \infty} \frac{1}{t_n} \int_{L-t_n}^L \int_0^1 u_x^2 dy dx \leq \lambda. \end{aligned}$$

Now that we know $|u_y^L| = 1$ a.e., we may extend u to $(0, \infty) \times (0, 1)$ by setting $u(x, y) = u^L(y)$ if $x \geq L$. This permits us to take $t < 0$ in the definition of the comparison functions u^t . Since u is independent of x for $x > L$, one has with $s = -t$

$$\begin{aligned} 0 &\leq I(u^t) - I(u) \\ &\leq \int_0^{(1+s)L} \int_0^1 \left[u_x^2(1+s) + \varepsilon |u_{yy}| \frac{1}{1+s} \right] dy dx \\ &\quad - \int_0^{(1+s)L} \int_0^1 u_x^2 + \varepsilon |u_{yy}| dy dx + sL \int_0^1 \varepsilon |u_{yy}^L| dy. \end{aligned}$$

Dividing by s and letting $s \rightarrow 0$ gives

$$\lambda \leq \varepsilon \int_0^1 |u_{yy}^L| dy.$$

Comparison with (2.15) shows that equality holds, and also that

$$\lim_{n \rightarrow \infty} \frac{1}{t_n} \int_{L-t_n}^L \int_0^1 u_x^2 dy dx = 0.$$

Since the latter holds for any sequence $t_n \rightarrow 0$, the proof is finished.

OBSERVATION. The statement of Lemma 2.4 demands that u be a minimizer of I^ε . The proof, however, depends primarily on taking first variations. Therefore it applies equally when u is a "stationary point" of I^ε — except for the estimate on λ , (2.9), which makes use of Lemma 2.3.

There is one more basic tool to present. It is a local version of the self-similar construction we used to prove Lemma 2.3. That lemma required a test function u such that $u = 0$ at $x = 0$; the value of u at the right-hand boundary $x = L$ was not

important. The following lemma interpolates similarly between *arbitrarily given* functions at the left-hand and right-hand boundaries.

LEMMA 2.5. *Fix $\ell > 0$. Let $u_0, u_1 \in W^{1,\infty}(0, 1)$ be given, with $|(u_0)'| \leq 1$ and $|(u_1)'| \leq 1$. Assume that $\varepsilon^{1/3}\ell^{2/3} \leq 1$. Then there exists $u_\varepsilon \in W^{1,2}((0, \ell) \times (0, 1))$ such that*

$$\begin{aligned} u_\varepsilon &= u_0 & \text{at} & \quad x = 0, \\ u_\varepsilon &= u_1 & \text{at} & \quad x = \ell, \\ |u_{\varepsilon y}| &= 1 & \text{a.e.} & \end{aligned}$$

and such that for any $\delta > 0$,

$$(2.16) \quad \int_0^\ell \int_0^1 u_{\varepsilon x}^2 + \varepsilon |u_{yy}|^2 dy dx \leq \frac{1 + \delta}{\ell} \int_0^1 |u_1(y) - u_0(y)|^2 dy + C_\delta \varepsilon^{2/3} \ell^{1/3}.$$

The constant C_δ depends on δ , but it is independent of u_0 , u_1 , ℓ , and ε .

OBSERVATION. The estimate (2.16) can be motivated as follows. When $\varepsilon = 0$ the minimum of the left-hand side is easily seen to be $\ell^{-1} \int |u_1(y) - u_0(y)|^2 dy$. We shall construct our test function for $\varepsilon > 0$ by superimposing oscillations upon the minimizer of the relaxed ($\varepsilon = 0$) problem. The estimate (2.16) asserts that the additional energy due to ε being positive is of exactly the same order as it was in Lemma 2.3.

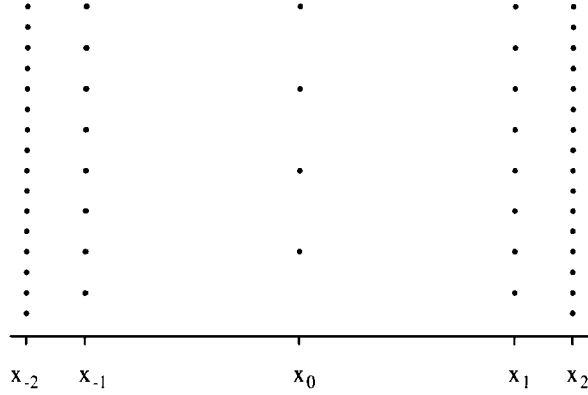
Proof: Let u_ℓ be the linear interpolation between u_0 and u_1 ,

$$u_\ell(x, y) = \frac{\ell - x}{\ell} u_0(y) + \frac{x}{\ell} u_1(y).$$

(This is the minimizer of the relaxed problem.) We divide up the interval $(0, \ell)$ geometrically: the subdivision points are

$$x_i = \ell - \frac{\ell}{2} \theta^i \text{ for } i = 0, 1, 2, \dots; \quad x_i = \frac{\ell}{2} \theta^{-i} \text{ for } i = 0, -1, -2, \dots.$$

We shall focus our attention on $x \geq \frac{\ell}{2}$, the case $x \leq \frac{\ell}{2}$ being symmetric. For $i = 0, 1, 2, \dots$, choose $N_i = 2^i N_0$ (N_0 to be chosen later) equidistant points on the segment $\{x_i\} \times (0, 1)$ (see Figure 4). Next, for each i choose a function v_i which agrees with u_ℓ at these equidistant points, and which satisfies $|(v_i)_y| = 1$ a.e. This is possible, since $|(u_\ell)_y| \leq 1$. Indeed we may choose v_i so that $(v_i)_y$ changes sign at most once between two successive points (see Figure 5). Finally, choose u_ε to be a piecewise linear interpolant: it is equal to v_i on $\{x_i\} \times (0, 1)$ for each i , and it satisfies $|u_{\varepsilon y}| = 1$ a.e. (One possible choice is indicated in Figure 6.)

Figure 4. Points where u_ε agrees with the linear interpolation u_ℓ .

It is easy to see that $|v_i - u_\ell(x_i, \cdot)| \rightarrow 0$ uniformly as $i \rightarrow \infty$, so

$$u_\varepsilon = u_1 \text{ at } x = \ell.$$

We have

$$\varepsilon \int_{x_i}^{x_{i+1}} \int_0^1 |u_{\varepsilon yy}| dy dx \leq C\varepsilon \Delta x N,$$

with $\Delta x = x_{i+1} - x_i$. On each of the triangles in Figure 6 we have

$$|u_{\varepsilon x}| \leq (|u_\varepsilon(x_{i+1}, y_k) - u_\varepsilon(x_i, y_k)| + 2\Delta y)(\Delta x)^{-1}.$$

Thus

$$\begin{aligned} & \int_{x_i}^{x_{i+1}} \int_0^1 |u_{\varepsilon x}|^2 dy dx \\ & \leq \sum_{k=1}^N \left\{ (1 + \delta) \frac{|u_\ell(x_{i+1}, y_k) - u_\ell(x_i, y_k)|^2}{(\Delta x)^2} + C_\delta \left(\frac{\Delta y}{\Delta x} \right)^2 \right\} \Delta y \Delta x \\ & \leq \frac{1 + \delta}{\ell^2} (\Delta x) N^{-1} \sum_{k=1}^N \left| u_1 \left(\frac{k}{N} \right) - u_0 \left(\frac{k}{N} \right) \right|^2 + C_\delta (\Delta x)^{-1} N^{-2}. \end{aligned}$$

Now, for any function f with $|f'| \leq 1$ we have the estimate

$$N^{-1} \sum_{k=1}^N \left| f \left(\frac{k}{N} \right) \right|^2 \leq (1 + \delta) \int_0^1 f^2(y) dy + C_\delta N^{-2}.$$

Redefining δ , it follows that

$$\begin{aligned} \int_{x_i}^{x_{i+1}} \int_0^1 |u_{\varepsilon x}|^2 dy dx & \leq (1 + \delta) \ell^{-2} \Delta x \int_0^1 |u_1 - u_0|^2 dy + C_\delta \ell^{-2} \Delta x N^{-2} \\ & \quad + C_\delta (\Delta x)^{-1} N^{-2}. \end{aligned}$$

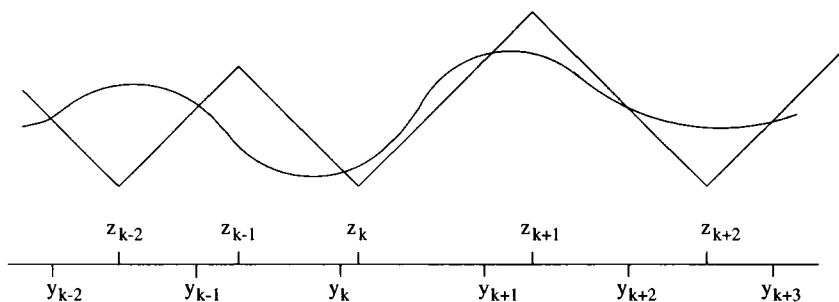


Figure 5. Approximation of a function with slope ≤ 1 by one with slope ± 1 . The two functions agree at evenly spaced points y_k . The slope of the approximating function changes from ± 1 to ∓ 1 at points z_k , with $y_k \leq z_k \leq y_{k+1}$.

Taking into account that $(\Delta x) = (\Delta x)_i = \frac{\ell}{2} \theta^i (1 - \theta)$ and $N = N_i = 2^i N_0$, we get

$$\begin{aligned} \int_0^\ell \int_0^1 u_{\varepsilon x}^2 + \varepsilon |u_{\varepsilon y y}| dy dx &= 2 \int_{\ell/2}^\ell \int_0^1 u_{\varepsilon x}^2 + \varepsilon |u_{\varepsilon y y}| dy dx \\ &\leq 2 \sum_{i=0}^\infty \frac{(1 + \delta)(\Delta x)_i}{\ell^2} \int_0^1 |u_1 - u_0|^2 dy + C\varepsilon \sum_{i=0}^\infty (\Delta x)_i N_i \\ &\quad + C\delta \sum_{i=0}^\infty (\ell^{-2} (\Delta x)_i N_i^{-2} + (\Delta x)_i^{-1} N_i^{-2}). \end{aligned}$$

Since $\sum_{i=0}^\infty (\Delta x)_i = \ell/2$, the first term is

$$\frac{1 + \delta}{\ell} \int_0^1 |u_1 - u_0|^2 dy,$$

as desired. Choosing $\frac{1}{2} < \theta < \frac{1}{4}$ as usual, the remaining terms are bounded by

$$C\delta \ell^{-1} N_0^{-2} + \varepsilon \ell N_0.$$

Finally, we take $N_0^{-1} \approx \varepsilon^{1/3} \ell^{2/3}$ to obtain (2.16).

We are now ready for the main result of this section. Recall that we proved $\min I^\varepsilon \leq C\varepsilon^{2/3} L^{1/3}$ in Lemma 2.3, by means of a self-similar construction. The question naturally arises whether this is the “right” scaling law, or whether some other construction could give a better result. Theorem 2.6 shows that it is indeed the right law, by proving the opposite inequality $\min I^\varepsilon \geq c\varepsilon^{2/3} L^{1/3}$. Actually, the theorem gives considerably more information: it shows that the energy of a minimizer scales similarly not only on the whole rectangle $0 < x < L$, but also on any subrectangle $0 < x < \ell$. Also, the lower bound we obtain is actually valid pointwise — it bounds the “energy at ℓ ” for any $\ell \in (0, L)$ (see (2.18)). We do not have a detailed description of the minimizers of I^ε . But Theorem 2.6 strongly suggests that they should resemble our self-similar construction.

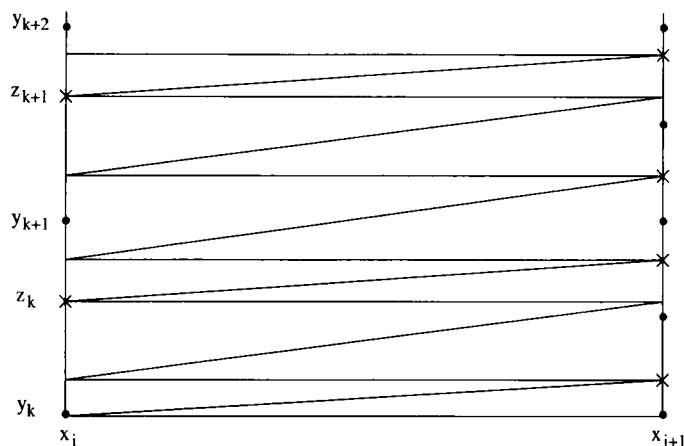


Figure 6. The function u_ε is linear on each triangle. It agrees with u_ℓ at the y_k 's (marked by dots); $u_{\varepsilon y}$ changes sign at the z_k 's (marked by x's). Notice that $u_y = \pm 1$ throughout the strip. This triangularization is obtained by drawing a horizontal line from each z_k on either side to divide the strip into rectangles, then subdividing each rectangle into two triangles. (It is but one of many possible choices.)

THEOREM 2.6. *Let u be any minimizer of I^ε . Then there exists a positive constant C such that for any $\ell \in (0, L)$*

$$(2.17) \quad \int_0^\ell \int_0^1 u_x^2 + \varepsilon |u_{yy}| dy dx \leq C \varepsilon^{2/3} \ell^{1/3};$$

also, there exists another constant c such that for a.e. $\ell \in (0, L)$

$$(2.18) \quad \int_0^1 u_x^2 + \varepsilon |u_{yy}|(\ell, y) dy \geq c \varepsilon^{2/3} \ell^{-2/3}.$$

The constants C and c are independent of ε , ℓ , L , and u . (As usual, however, it is assumed that $\varepsilon^{1/3} L^{2/3} \ll 1$.)

OBSERVATION 1. Integration of (2.18) from 0 to L gives $\min I^\varepsilon \geq 3c \varepsilon^{2/3} L^{1/3}$. We shall give a quantitative estimate for the constant c in Lemma 2.8.

OBSERVATION 2. By equipartition of energy (Lemma 2.4), inequality (2.18) holds for each term in the energy (u_x^2 and $\varepsilon |u_{yy}|$) individually, if $\ell \ll L$. Thus the average width of the “twins” (the regions of constant u_y) is $\leq C \varepsilon^{1/3} \ell^{2/3}$ for “most” values of ℓ .

OBSERVATION 3. These results are consistent with the self-similar construction used to prove Lemma 2.3, provided that in doing the construction one chooses $\theta = 2^{-3/2}$. Based on that construction and the scale invariance of the energy, it is

tempting to conjecture a local-in- y scaling law of the form

$$\int_{Q_\ell} u_x^2 + \varepsilon |u_{yy}| \, dy dx \leq C \varepsilon^{2/3} \ell ,$$

with $Q_\ell = (0, \ell) \times (a, a + \ell^{2/3})$ for any $a \in (0, 1)$. We have thus far been unable to prove such an estimate. We do have some related but weaker results; they will be presented in [19].

OBSERVATION 4. It is plausible that there should be an upper bound analogous to (2.18). We have again been unable to prove such an estimate.

Proof of Theorem 2.6: Our first goal is the upper bound (2.17). We shall use a standard comparison argument, with a slight twist involving the equipartition of energy. Let u be a minimizer of I^ε . For any $\ell \in (0, L)$, consider the competitor

$$\tilde{u} = \begin{cases} u & \text{for } x > \ell , \\ u_\varepsilon & \text{for } x \leq \ell , \end{cases}$$

where u_ε is as in Lemma 2.5 with $u_0 = 0, u_1 = u|_{x=\ell}$. Let $R_\ell = (0, \ell) \times (0, 1)$, and notice the Poincaré inequality

$$(2.19) \quad \frac{1}{\ell} \int_0^1 u^2(\ell, y) \, dy \leq \int_{R_\ell} u_x^2 \, dx dy .$$

Since

$$0 \leq I^\varepsilon(\tilde{u}) - I^\varepsilon(u) = \int_{R_\ell} u_{\varepsilon x}^2 + \varepsilon |u_{\varepsilon yy}| \, dx dy - \int_{R_\ell} u_x^2 + \varepsilon |u_{yy}| \, dx dy ,$$

we deduce using Lemma 2.5 and (2.19) that

$$(2.20) \quad 0 \leq (1 + \delta) \int_{R_\ell} u_x^2 \, dx dy + C_\delta \varepsilon^{2/3} \ell^{1/3} - \int_{R_\ell} u_x^2 + \varepsilon |u_{yy}| \, dx dy .$$

At this point we seem to be stuck: the standard idea would be to absorb the positive term $(1 + \delta) \int u_x^2$ into the negative term $\int u_x^2 + \varepsilon |u_{yy}|$, but we would appear to have $\delta \int u_x^2$ left over. To overcome this difficulty, we make use of equipartition (Lemma 2.4). It shows that

$$\begin{aligned} \int_{R_\ell} \varepsilon |u_{yy}| \, dx dy &= \int_{R_\ell} u_x^2 - \ell \lambda \\ &\geq \int_{R_\ell} u_x^2 - C \varepsilon^{2/3} \ell^{1/3} . \end{aligned}$$

Combining this with (2.20) for $\delta = \frac{1}{2}$ we obtain the desired upper bound for $\int_{R_\ell} u_x^2$. A second application of equipartition gives a similar bound for $\int_{R_\ell} \varepsilon |u_{yy}|$, completing the proof of (2.17).

We turn now to the lower bound (2.18). The basic idea is as follows. Suppose that the restriction of the energy to the vertical segment $x = \ell$ were very small. Focusing attention on the second term $\int \varepsilon |u_{yy}| dy$, it would follow that u_y did not change sign many times along the vertical line $x = \ell$. So the restriction of u to this segment would be a sawtooth function with slopes ± 1 and a restricted number of corners. Such a function must have a relatively large L^2 norm. But then we would have $u = 0$ at $x = 0$ and $\|u\|_{L^2}$ large at $x = \ell$. This would entail a large value of $\int u_x^2 dx dy$, contradicting the upper bound (2.17).

We now make this argument quantitative. The crucial estimate “small $\int |u_{yy}|$ implies large $\int u^2$ ” is proved as Lemma 2.7 below. It shows that $u_1 = u|_{x=\ell}$ satisfies an estimate of the form

$$\int_0^1 u_1^2 dy \geq c \left(\int_0^1 |u_{1yy}| dy + 1 \right)^{-2}.$$

By the Poincaré inequality (2.19) and the upper bound (2.17) we have that

$$\int_0^1 u_1^2 dy \leq \ell \int_{R_\ell} u_x^2 dx dy \leq C \varepsilon^{2/3} \ell^{4/3}.$$

It follows that

$$\int_0^1 |u_{yy}|(\ell, y) dy \geq c \varepsilon^{-1/3} \ell^{-2/3} - 1.$$

This implies the lower bound (2.18).

We made use above of

LEMMA 2.7. *Let $f \in W^{1,\infty}(0, 1)$. Assume that $|f'| = 1$ a.e. and that f' changes sign N times. Then*

$$\int_0^1 f^2 dx \geq \frac{1}{12} (N+1)^{-2} = \frac{1}{12} \left(\frac{1}{2} \int_0^1 |f''| dx + 1 \right)^{-2}.$$

Proof: Let $x_1 < \dots < x_N$ be the points where f' changes sign, and set $x_0 = 0$, $x_{N+1} = 1$. Then, for $i = 0, \dots, N$, we have

$$\int_{x_i}^{x_{i+1}} f^2(y) dy = \left(\frac{f(x_i) + f(x_{i+1})}{2} \right)^2 (x_{i+1} - x_i) + \frac{1}{12} (x_{i+1} - x_i)^3.$$

It follows, using the convexity of $t \mapsto t^3$, that

$$\int_0^1 f^2(y) dy \geq \sum_{i=0}^N \frac{1}{12} (x_{i+1} - x_i)^3 \geq \frac{1}{12} (N+1)^{-2}.$$

We remark that this estimate is sharp: it is achieved when f has mean value zero and points x_N are equally spaced, so that the graph of f is a periodic sawtooth.

For the application to twin branching (see [18]), it is useful to have good quantitative estimates for the constants in our scaling laws for $\min I^\varepsilon$. To estimate the value of C in our upper bound $\min I^\varepsilon \leq C\varepsilon^{2/3}L^{1/3}$, it suffices to examine carefully the self-similar construction used for Lemma 2.3. This is done in [18]; it leads to the conclusion that $C \leq 6.86$. To estimate the value of c in our lower bound $\min I^\varepsilon \geq c\varepsilon^{2/3}L^{1/3}$, one option would be to repeat the proof of (2.18) with more careful attention to the arithmetic. We shall do a bit better, however, by arguing differently:

LEMMA 2.8. *The estimate*

$$\min\{I^\varepsilon(u) : u \in \mathcal{A}_0\} \geq c\varepsilon^{2/3}L^{1/3}$$

holds with

$$c = (3 - c'\varepsilon^{1/3}L^{2/3})^{2/3}.$$

OBSERVATION. The reader is reminded that we always assume $\varepsilon^{1/3}L^{2/3} \ll 1$. So Lemma 2.8 asserts that c is essentially $3^{2/3} \approx 2.08$. Thus the constant in our upper bound and that in our lower bound differ from one another by roughly a factor of 3. Our earlier paper [18] makes reference to a lower bound with constant $c = (9/4)^{1/3} \approx 1.31$; the result presented here is obviously better.

Proof: Let u be a minimizer of I^ε , and let

$$B_1 = \int_{R_L} \varepsilon |u_{yy}| \, dx dy, \quad B_2 = \int_{R_L} u_x^2 \, dx dy, \quad B = B_1 + B_2, \quad \mu = \frac{B_1}{B}.$$

For any ℓ , $0 < \ell < L$, we have

$$(2.21) \quad B_2 \geq \int_{R_\ell} u_x^2 \, dx dy \geq \frac{1}{\ell} \int_0^1 u^2(\ell, y) \, dy,$$

making use of the Poincaré inequality (2.19). Let $N(\ell)$ be the number of times u_y changes sign along the line $x = \ell$, which is well-defined for almost every ℓ :

$$N(\ell) = \frac{1}{2} \int_0^1 |u_{yy}(\ell, y)| \, dy.$$

Combining (2.21) with Lemma 2.7 we get

$$(N(\ell) + 1)B_2^{1/2} \geq \frac{1}{\sqrt{12}} \ell^{-1/2}.$$

Integrating this from 0 to L yields

$$\left(\frac{1}{2\varepsilon}B_1 + L\right)B_2^{1/2} \geq \frac{1}{\sqrt{3}}L^{1/2},$$

or equivalently

$$\frac{1}{2\varepsilon}\mu(1-\mu)^{1/2}B^{3/2} + L(1-\mu)^{1/2}B^{1/2} \geq \frac{1}{\sqrt{3}}L^{1/2}.$$

This is most conveniently expressed in terms of $B' = \varepsilon^{-2/3}L^{-1/3}B$: it says that

$$\frac{1}{2}\mu(1-\mu)^{1/2}(B')^{3/2} + \varepsilon^{1/3}L^{2/3}(1-\mu)^{1/2}(B')^{1/2} \geq \frac{1}{\sqrt{3}}.$$

We know that B' is of order 1, so it follows that

$$\mu(1-\mu)^{1/2}(B')^{3/2} \geq \frac{2}{\sqrt{3}} - C\varepsilon^{1/3}L^{2/3}.$$

The worst choice for μ is $\mu = \frac{2}{3}$ which gives $\mu(1-\mu)^{1/2} = 2/(3\sqrt{3})$, so

$$(B')^{3/2} \geq 3 - c'\varepsilon^{1/3}L^{2/3}.$$

The lemma follows.

3. The Model Energy with a Soft Boundary Condition

This section addresses the “soft boundary condition” version of our model energy. Physically, this corresponds to treating the austenite as an elastic material rather than as a rigid substrate. Mathematically, it has the effect of replacing the boundary condition $u = 0$ at $x = 0$ by a penalization term involving the $H^{1/2}$ norm of $u|_{x=0}$. There is a penalization parameter β , which represents the rigidity of the austenite. (More precisely, β is the ratio between the rigidity of the austenite and that of the martensite, so in most cases it will be a parameter of order 1.) If β is not too small — specifically, if $\beta \gg (\varepsilon/L)^{1/3}$ — then the situation is much like that of the “hard” boundary condition. If β is sufficiently small, however — namely, if $\beta \ll (\varepsilon/L)^{1/3}$, then the situation is different. In this regime an essentially one-dimensional test function has lower energy than our self-similar construction.

It is convenient (though not essential) to work with y -periodic functions, since this permits us to define the $H^{1/2}$ norm via Fourier series. So in this section we impose the condition

$$(3.1) \quad u(x, 0) = u(x, 1) \quad \text{for every } x \in (0, L).$$

Strictly speaking u should be viewed as a function on the cylinder $Z = (0, L) \times (\mathbb{R}/\mathbb{Z})$. In particular, the expression $\int \varepsilon |u_{yy}| dx dy$ must be understood as counting

the contribution of any jumps of u_y along the top/bottom boundary. Begging the reader's indulgence, we shall abuse notation slightly by continuing to write $R_L = (0, L) \times (0, 1)$ rather than Z as the domain of integration.

We now formulate the "soft boundary condition" version of the problem more precisely. The functional to be minimized is

$$(3.2) \quad J^\varepsilon(u) = \int_{R_L} u_x^2 + \varepsilon |u_{yy}| \, dx dy + \beta \|u_0\|_{1/2}^2$$

subject to the constraint $|u_y| = 1$ and the periodicity boundary condition (3.1). Here $u_0 = u|_{x=0}$ (in the sense of trace). The last term in J^ε is defined by

$$\|u_0\|_{1/2}^2 = \sum_{k=-\infty}^{\infty} |k| |\hat{u}_0(k)|^2,$$

where $\hat{u}_0(k)$ are the Fourier coefficients of u_0 , i.e.,

$$u_0(y) = \sum_{k=-\infty}^{\infty} \hat{u}_0(k) e^{2\pi i k y}.$$

As explained in the Introduction, there is also an alternative characterization

$$\|u_0\|_{1/2}^2 = \min_{v=u_0 \text{ at } x=0} \frac{1}{2\pi} \int_{-\infty}^0 \int_0^1 (v_x^2 + v_y^2) \, dx dy.$$

We are abusing notation a bit here: this is not really a norm on the space $H^{1/2}$, but rather on $H^{1/2}/\mathbb{R}$, since by the above definition constant functions have norm zero. (See the Appendix for more information on the space $H^{1/2}$ and related matters.)

The precise class of admissible functions for J^ε is

$$\mathcal{A} = \{u \in H^1(Z) : |u_y| = 1 \text{ a.e. and } u_{yy} \text{ is a Radon measure with finite mass}\}.$$

(Recall that Z denotes the cylinder, reflecting the y -periodicity of u .) We continue to assume that $\varepsilon^{1/3} L^{2/3} \ll 1$.

The existence theory is parallel to that for the hard boundary condition.

THEOREM 3.1. *For $\varepsilon > 0$, J^ε attains its minimum on $\mathcal{A}$.*

Proof: This follows from the proof of Theorem 2.1, taking into account that $u^{(j)} \rightharpoonup u$ in $H^1(R_L)$ implies $u_0^{(j)} \rightharpoonup u_0$ in $H^{1/2}(0, 1)$ (see the Appendix, especially (A.5)) and that $\|\cdot\|_{1/2}^0$ is weakly lower semicontinuous.

Every test function for the "hard boundary condition" is automatically a test function for the "soft boundary condition." Hence we immediately deduce an upper bound for the minimum of J^ε from the analysis of Section 2. Throughout

the following, all constants are understood to be independent of both β and ϵ (unless otherwise specified).

LEMMA 3.2. *There is a constant C such that*

$$(3.3) \quad \min\{J^\epsilon(u) : u \in \mathcal{A}\} \leq C\epsilon^{2/3}L^{1/3}.$$

Proof: This follows directly from the self-similar construction used to prove Lemma 2.3. (The construction can be done so as to respect the condition of y -periodicity, (3.1).)

The principle of “equipartition of energy” applies to J^ϵ exactly as to I^ϵ . We omit the assertions about u^L for the sake of brevity.

LEMMA 3.3. *Let u be a minimizer of J^ϵ on $\mathcal{A}$. Then there exists a constant λ such that*

$$(3.4) \quad \int_0^1 \epsilon |u_{yy}|(x, y) dy - \int_0^1 u_x^2(x, y) dy = \lambda.$$

The value of λ depends on u and hence implicitly on ϵ , L , and β ; however it satisfies an estimate of the form

$$(3.5) \quad |\lambda| \leq C\epsilon^{2/3}L^{-2/3}.$$

Proof: The proof of Lemma 2.4 applies *verbatim*, since it makes use of variations that do not alter u_0 .

In Theorem 2.6 we established upper and lower bounds for the energy in any rectangle $R_\ell = (0, \ell) \times (0, 1)$, when u is a *minimizer*. The upper bound extends easily to the soft boundary condition.

THEOREM 3.4. *There is a constant C such that for any minimizer u of J^ϵ and for any $\ell \in (0, L)$:*

$$\int_0^\ell \int_0^1 u_x^2 + \epsilon |u_{yy}| dy dx \leq C\epsilon^{2/3}\ell^{1/3}.$$

Proof: The argument is closely parallel to that of Theorem 2.6. One must choose u_ϵ as in Lemma 2.5 with $u_0 = u|_{x=0}$, $u_1 = u|_{x=\ell}$. The Poincaré inequality (2.19) is replaced by

$$\frac{1}{\ell} \int_0^1 |u_1 - u_0|^2 dy \leq \int_{R_\ell} u_x^2 dx dy.$$

Further details can safely be left to the reader.

Proving lower bounds for J^ε turns out to be a more subtle matter. The reason is that *there can be no bound of the form* $\min\{J^\varepsilon\} \geq c\varepsilon^{2/3}L^{1/3}$ *with* c *independent of* β . To see this, consider a test function v for J^ε which depends only on y : $v(x, y) = v_0(y)$. Choosing v_0 to be a periodic sawtooth with optimally chosen period, one obtains $J^\varepsilon(v) \approx c'\varepsilon^{1/2}\beta^{1/2}L^{1/2}$ (see, e.g., [18]). Thus a lower bound of the form $c\varepsilon^{2/3}L^{1/3}$ can only be valid if β is of order $(\varepsilon/L)^{1/3}$ or larger. We conjecture that the right bound is

$$(3.6) \quad \min J^\varepsilon \geq c \min\{\varepsilon^{1/2}\beta^{1/2}L^{1/2}; \varepsilon^{2/3}L^{1/3}\}.$$

While we are not able to prove (3.6), we are able to prove the analogous statement for a slightly modified functional. The modification consists of replacing the square of the $H^{1/2}$ norm by the L^2 norm (to the first power). To motivate this, we recall the interpolation inequality (see (A.6))

$$\|u_0\|_{1/2}^2 \leq \frac{1}{2\pi} \|u_0\|_{L^2} \|u'_0\|_{L^2}.$$

Here as usual, $u_0(x) = u|_{x=0}$. It is tempting, but wrong, to think that $|u'_0| = 1$ a.e. (After all, the self-similar construction achieves $u_0 = 0$.) But it is true that $|u'_0| \leq 1$. To see this, we note that $u(x, \cdot)$ converges to $u_0(\cdot)$ in L^2 as $x \rightarrow 0$ (by the Poincaré inequality) and $|u_y(x, \cdot)| \leq 1$. Hence $u_y(x, \cdot) \xrightarrow{*} u'_0(\cdot)$ in $L^\infty(0, 1)$ and $|u'_0| \leq 1$. Combining this observation with the interpolation inequality, we deduce that

$$\|u_0\|_{1/2}^2 \leq \frac{1}{2\pi} \|u_0\|_{L^2}.$$

We expect that for the minimizer u of J^ε , u_0 looks more or less like a periodic sawtooth with slope ± 1 . In that case one would also have the reverse estimate

$$\|u_0\|_{1/2}^2 \geq c \|u_0\|_{L^2} \|u'_0\|_{L^2} = c \|u_0\|_{L^2}.$$

With this motivation, we now consider the modified functional

$$\tilde{J}^\varepsilon(u) = \int_{R_L} u_x^2 + \varepsilon |u_{yy}| dx dy + \frac{\beta}{2\pi} \|u_0\|_{L^2}.$$

One easily verifies that $\tilde{J}^\varepsilon$ attains its minimum on $\mathcal{A}$, that a global upper bound of the form (3.3) holds, and that minimizers have “equipartioned energy” in the sense of (3.4)–(3.5). We also have the following result, which yields as a corollary the analogue of (3.6). It suggests (though it does not prove) that the minimizers of $\tilde{J}^\varepsilon$ are essentially one dimensional when $\beta \ll (\varepsilon/L)^{1/3}$. It further suggests that the minimizers resemble our self-similar construction when $\beta \gg (\varepsilon/L)^{1/3}$, except in a layer of width approximately $\varepsilon\beta^{-3}$ near the left-hand boundary $x = 0$.

THEOREM 3.5. *There exist positive constants C, c such that for any minimizer u of $\tilde{J}^\varepsilon$ the following hold:*

(i) for any $\ell \in (0, L)$,

$$(3.7) \quad \int_0^\ell \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx \leq C \varepsilon^{2/3} \ell^{1/3} ;$$

(ii) if $\beta \geq (\varepsilon/L)^{1/3}$, then

$$(3.8) \quad \|u_0\|_{L^2} \leq C \varepsilon \beta^{-2} ,$$

and for a.e. $\ell \in (0, L)$

$$(3.9) \quad \int_0^1 \varepsilon |u_{yy}|(\ell, y) \, dy \geq c \varepsilon^{2/3} (\ell^{2/3} + \varepsilon^{2/3} \beta^{-2})^{-1} ;$$

(iii) if $\beta \leq (\varepsilon/L)^{1/3}$ (and $\beta \gg \varepsilon/L$) then

$$(3.10) \quad \|u_0\|_{L^2} \leq C \varepsilon^{1/2} L^{1/2} \beta^{-1/2} ,$$

and for a.e. $\ell \in (0, L)$

$$(3.11) \quad \int_0^1 \varepsilon |u_{yy}|(\ell, y) \, dy \geq c \varepsilon^{1/2} L^{-1/2} \beta^{1/2} .$$

Proof: The proof of (i) is exactly the same as that of Theorem 3.4. Turning to (ii), suppose that $\beta > (\varepsilon/L)^{1/3}$. To prove the bound (3.8) on $\|u_0\|_{L^2}$, we introduce a competitor $\tilde{u}$ obtained as follows. First, using Lemma 2.5, extend u to a strip $-d < x < 0$, with $u = 0$ at $x = -d$. (The value of the parameter d will be chosen later.) Then translate the extended u to the right to get $\tilde{u}$, i.e., set $\tilde{u}(x, y) = u(x + d, y)$. The inequality $\tilde{J}^\varepsilon(u) \leq \tilde{J}^\varepsilon(\tilde{u})$ gives

$$\begin{aligned} \tilde{J}^\varepsilon(u) &\leq \int_{-d}^0 \int_0^1 u_{\varepsilon x}^2 + \varepsilon |u_{\varepsilon yy}| \, dy dx + \int_0^{L-d} \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx \\ &\leq \frac{1+\delta}{d} \int_0^1 u_0^2 \, dy + C_\delta \varepsilon^{2/3} d^{1/3} + \int_0^L \int_0^1 u_x^2 + \varepsilon |u_{yy}| \, dy dx . \end{aligned}$$

It follows that

$$\frac{\beta}{2\pi} \|u_0\|_{L^2} \leq \frac{1+\delta}{d} \|u_0\|_{L^2}^2 + C_\delta \varepsilon^{2/3} d^{1/3} .$$

If we choose $\delta = 1$ and $d = \frac{8\pi}{\beta} \|u_0\|_{L^2}$, then the estimate becomes

$$\beta \|u_0\|_{L^2} \leq C \varepsilon^{2/3} \beta^{-1/3} \|u_0\|_{L^2}^{1/3} ,$$

which is equivalent to (3.8).

We turn next to the lower bound in (ii), (3.9). Let $u = u|_{x=\ell}$ and (as usual) $u_0 = u|_{x=0}$. From (3.7) and the Poincaré inequality we have

$$(3.12) \quad \|u_\ell - u_0\|_{L^2}^2 \leq C \varepsilon^{2/3} \ell^{4/3} .$$

Hence by the triangle inequality

$$\|u_\ell\|_{L^2} \leq C\epsilon^{1/3}\ell^{2/3} + \|u_0\|_{L^2} \leq C(\epsilon^{1/3}\ell^{2/3} + \epsilon\beta^{-2}) .$$

By Lemma 2.7 this implies

$$\begin{aligned} \frac{1}{2} \int_0^1 \epsilon |u_{yy}|(\ell, y) dy &\geq c\epsilon(\epsilon^{1/3}\ell^{2/3} + \epsilon\beta^{-2})^{-1} - 1 \\ &\geq c'\epsilon^{2/3}(\ell^{2/3} + \epsilon^{2/3}\beta^{-2})^{-1} . \end{aligned}$$

Here we have used $\epsilon^{1/3}L^{2/3} \ll 1$ to know that $\epsilon\beta^{-2} = \epsilon^{1/3}L^{2/3}(\epsilon/L)^{2/3}\beta^{-2} \ll 1$.

Turning to (iii), we suppose now that $\epsilon/L \ll \beta \leq (\epsilon/L)^{1/3}$. The bound on $\|u_0\|_{L^2}$ is an easy consequence of the global upper bound

$$(3.13) \quad \min\{\tilde{J}^\epsilon(u) : u \in \mathcal{A}\} \leq C\epsilon^{1/2}\beta^{1/2}L^{1/2} .$$

This is proved considering one-dimensional test functions $u(x, y) = v(y)$, taking v to be a periodic sawtooth with slope ± 1 and period of order $\epsilon^{1/2}\beta^{-1/2}L^{1/2}$.

For the lower bound (3.11), we return to (3.12) (which is valid for all β). Taking into account that

$$\epsilon^{1/3}L^{2/3} \leq \epsilon^{1/2}\beta^{-1/2}L^{1/2}(\epsilon/L)^{-1/6}\beta^{1/2} \leq \epsilon^{1/2}L^{1/2}\beta^{-1/2} ,$$

we have

$$\|u|_{x=\ell}\|_{L^2} \leq C\epsilon^{1/3}\ell^{2/3} + \|u_0\|_{L^2} \leq C\epsilon^{1/2}\beta^{-1/2}L^{1/2} \ll 1 .$$

So (3.11) follows from Lemma 2.7.

COROLLARY 3.6. *There exist constants c, C such that*

$$c \min\{\epsilon^{1/2}\beta^{1/2}L^{1/2}; \epsilon^{2/3}L^{1/3}\} \leq \min_{\mathcal{A}} \tilde{J}^\epsilon \leq C \min\{\epsilon^{1/2}\beta^{1/2}L^{1/2}; \epsilon^{2/3}L^{1/3}\} .$$

Proof: The lower bound follows by integration from (3.9) and (3.11). The upper bound follows from Lemma 3.2 and (3.13).

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Appendix

In Section 3 we have made use of the space $H^{1/2}$, for periodic functions on $(0, 1)$. Here is a brief review of some relevant facts.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be periodic with period 1. Its Fourier expansion is

$$(A.1) \quad f(x) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i k x}.$$

For any real number $s \geq 0$, we say $f \in H^s$ if

$$(A.2) \quad \|f\|_{H^s} := \left(\sum_{k=-\infty}^{\infty} (1 + |2\pi k|^2)^s |\hat{f}(k)|^2 \right)^{1/2} < \infty.$$

The space H^s has other, equivalent characterizations using interpolation and via the integrability of certain “fractional difference quotients” — see, e.g., [1], [23], and the references therein. Note that for integer values of s the definition gives precisely the usual Sobolev spaces. The space $H^{1/2}$ plays a special role in our analysis because H^1 functions have boundary values in $H^{1/2}$ (see, e.g., [23]).

Actually, we are mainly interested in the space $H^{1/2}/\mathbb{R}$ obtained by dividing out the constant functions (e.g., by requiring $\int f = 0$). It is easy to see that

$$(A.3) \quad \|f\|_{1/2} := \left(\sum_{k=-\infty}^{\infty} |k| |\hat{f}(k)|^2 \right)^{1/2}$$

is an equivalent norm on $H^{1/2}/\mathbb{R}$. We claim that

$$(A.4) \quad \min_{\substack{u=f \text{ at } x=0 \\ u(x,0)=u(x,1)}} \int_{-\infty}^0 \int_0^1 u_x^2 + u_y^2 dy dx = 2\pi \|f\|_{1/2}^2.$$

Indeed, the minimizer has to satisfy

$$\Delta u = 0.$$

Since u is y -periodic, we can express it as

$$u(x, y) = \sum_{k=-\infty}^{\infty} g_k(x) e^{2\pi i k y}.$$

This gives

$$\begin{aligned} (g_k)_{xx} - (2\pi k)^2 g_k &= 0, \\ g_k(0) &= \hat{f}(k), \end{aligned}$$

whence

$$g_k(x) = e^{2\pi |k|x} \hat{f}(k).$$

It follows that

$$\begin{aligned} \int_{-\infty}^0 \int_0^1 u_x^2 + u_y^2 dx dy &= \sum_{k=-\infty}^{\infty} \int_{-\infty}^0 2e^{4\pi|k|x} (2\pi k)^2 |\hat{f}(k)|^2 dx \\ &= \sum_{k=-\infty}^{\infty} 2\pi |k| |\hat{f}(k)|^2 = 2\pi \|f\|_{1/2}^2. \end{aligned}$$

A similar argument shows that the map $u \mapsto u|_{x=0}$ is a continuous map from $H^1(R_L)$ to $H^{1/2}(0, 1)$. Since this map is linear, it follows that

$$(A.5) \quad u_k \rightharpoonup u \text{ in } H^1(R_L) \text{ implies } u_k|_{x=0} \rightharpoonup u|_{x=0} \text{ in } H^{1/2}(0, 1).$$

Finally we recall the interpolation inequality

$$(A.6) \quad \|f\|_{1/2}^2 \leq \frac{1}{2\pi} \|f\|_{L^2} \|f'\|_{L^2},$$

which follows directly from (A.2), (A.3), and the Cauchy-Schwarz inequality. One can see that the interpolation inequality is sharp up to a constant whenever the Fourier coefficients $\hat{f}(k)$ are more or less concentrated in an interval of the form (k_0, Ck_0) for some k_0 .

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